

Patient-Level Cardiac MRI Analysis with Out-of-Fold Heart Segmentation and Shortcut-Learning Controls

A technical narrative of how 63,425 MRI images became a leakage-controlled 30-patient experiment

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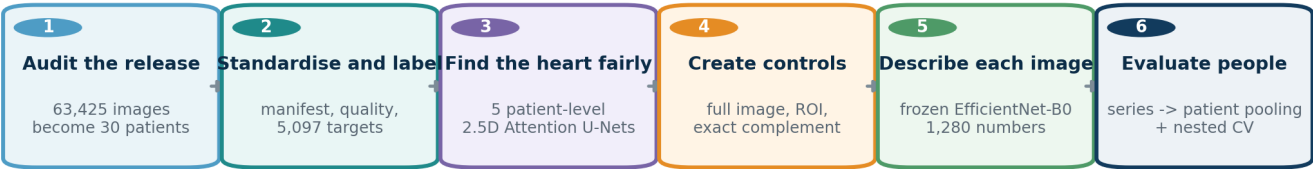
Scope. This is a methodological proof-of-concept using the public release's Normal/Sick labels. It is not a medical device, does not independently diagnose coronary artery disease, and has not been externally or clinically validated.

63,425 JPEG images	30 patients	2,859 series proxies	5,097 accepted targets	5 OOF models	0.938 primary ROI AUC
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Research question

When the same cardiac MRI slices are represented as the complete image, an automatically localised heart region, or the pixels outside that region, where is the label-related information most concentrated?

The pipeline as a scientific story



Central question: does the label-related signal remain strongest when the model is forced to focus on the heart?

Figure 1. The complete pipeline is organised as a sequence of methodological questions, not merely a sequence of models.

Main conclusion. Inside this small public release, the out-of-fold cardiac ROI separated the supplied labels more strongly than the same-slice full image or complement. The complement remained predictive, so dataset-specific shortcuts were reduced but not eliminated.

1. The dataset audit changed the problem

The first important model was a folder tree: it revealed that the apparent sample size was 30 people, not 63,425 independent images.

At the beginning, the release looked large enough for conventional image classification. A structural audit changed that interpretation. The 63,425 JPEG files were nested under only 30 top-level Directory_* folders, and the dataset creator confirmed that each top-level folder represented one patient. This meant that neighbouring slices, repeated frames, scanner style, and patient-specific anatomy could easily leak across an image-level split.

The first correction: define the independent unit before training

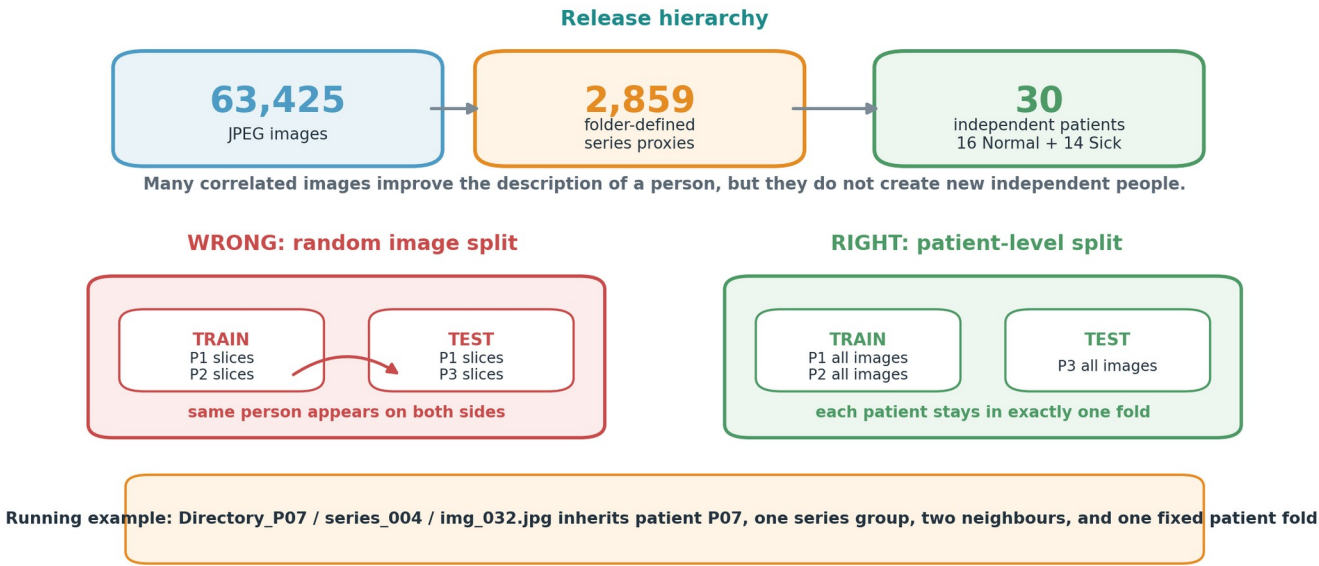


Figure 2. The hierarchy of the release and the difference between an image-level split and a patient-level split.

Why this matters. A classifier can score highly by recognising a person, acquisition pattern, or export style. Keeping every image from one patient in one fold makes the final test ask a harder question: does the method transfer to a person it did not use for fitting?

Audit item	Result used by the pipeline
Independent unit	Top-level Directory_* patient folder
Class balance	16 Normal patients; 14 Sick patients
Series structure	2,859 folder-defined proxies, not verified DICOM SeriesInstanceUIDs
Duplicate audit	1,094 exact duplicate groups; no exact cross-patient or cross-label group
Fixed fold principle	All slices, series, masks, tuning decisions, and final predictions remain grouped by patient

2. Stage 1 - Initial CPU: organise, standardise, and review

Before the neural network sees an image, the pipeline makes the data geometry, identifiers, quality state, and manual targets explicit.

Each discovered file becomes a manifest row with a stable image token, patient ID, series proxy, sequence position, neighbouring-frame paths, class label, and patient fold. Stable tokens are based on the path beginning at Directory_* so that masks and audit rows remain linked even if Kaggle mounts the dataset under a different root.

Stage 1 - Initial CPU: make every image traceable, comparable, and reviewable

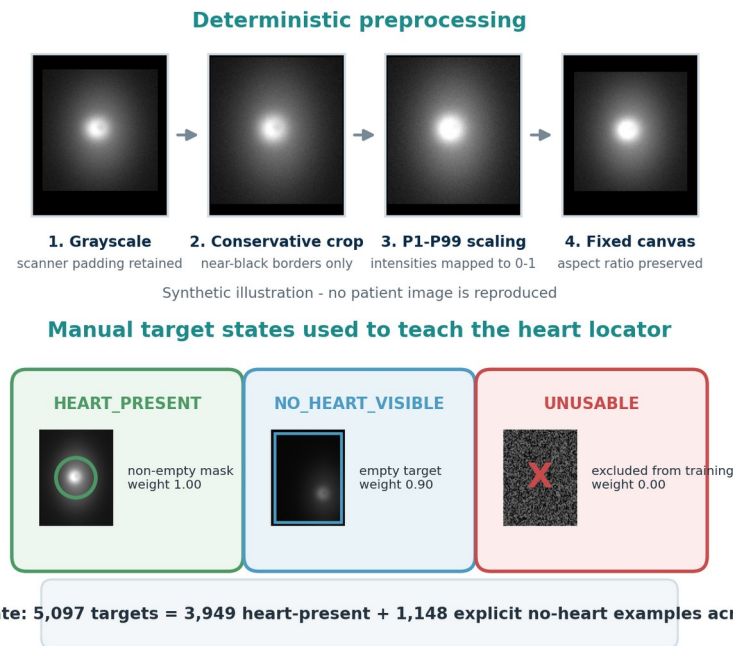


Figure 3. Deterministic preprocessing and the three semantic target states. All image examples are synthetic.

Pipeline component	Technical role
Preprocessing	Read grayscale; conservatively remove near-black padding; clip to the 1st/99th percentiles; preserve aspect ratio; centre on a 256 x 256 canvas; create a 224 x 224 classifier view.
Quality audit	Sharpness, noise ratio, dynamic range, and perceptual hash are measured for every image; thresholds combine fixed limits with per-patient tail quantiles.
Manual target semantics	HEART_PRESENT requires a non-trivial mask; NO_HEART_VISIBLE requires an empty target; UNUSABLE is excluded.
Integrity gates	At least 800 valid positive masks, positive masks in every fold, at least six patients with positives, and consistency between semantic labels and PNG content.
Accepted state	5,097 training targets: 3,949 heart-present and 1,148 no-heart; 161 rejected and 1,121 explicitly unusable.

Running example. The illustrative file Directory_P07/series_004/img_032.jpg becomes a 256 x 256 robust-scaled image, a 3-channel previous/current/next stack, and a target record linked by one stable token. It never becomes an independent patient.

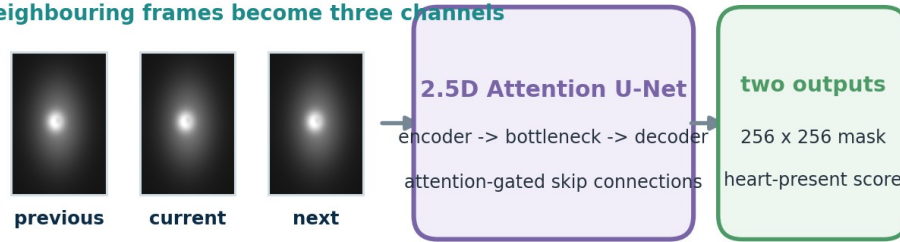
3. Stage 2 - GPU: learn a heart locator, not a disease classifier

The segmentation model receives no Normal/Sick label. Its job is to propose the general heart region and decide whether a mask should exist.

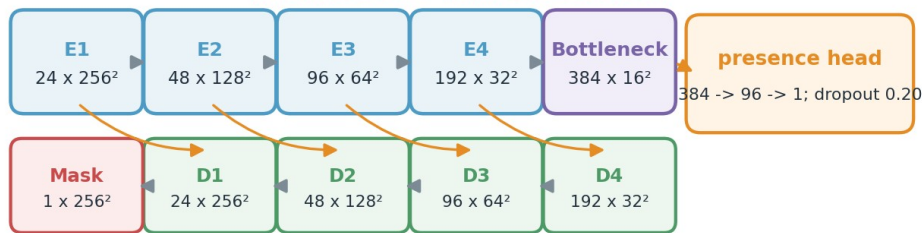
The locator is a 2.5D Attention U-Net. "2.5D" means that the previous, current, and next frames are stacked as three channels, giving local sequence context without assuming a complete DICOM volume. The network has an encoder, a 384-channel bottleneck, a decoder, attention-gated skip connections, one pixel-level mask output, and one auxiliary heart-presence output.

Stage 2 - GPU: learn to localise the heart, not to classify disease

three neighbouring frames become three channels



Architecture and tensor flow



Training protocol

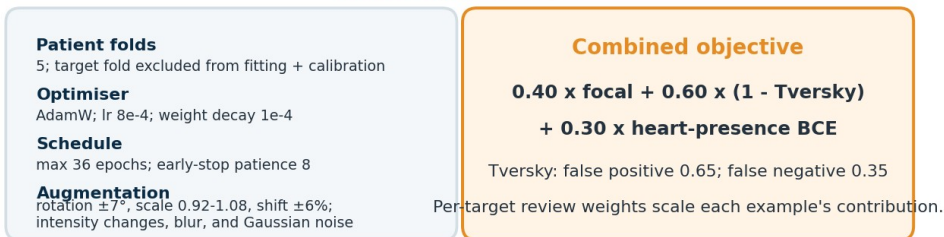


Figure 4. The 2.5D input, Attention U-Net tensor flow, and the training objective used in each patient fold.

Scientific separation. The segmentation target says where the heart is visible. It does not say whether the patient is Normal or Sick. This separation prevents the first model from directly optimising the final classification label.

Model-selection rule

For each target fold, every patient in that fold is excluded from segmentation fitting and threshold calibration. A validation-patient subset is selected only from the remaining patients. The best checkpoint is chosen from patient-balanced segmentation quality and heart-presence balanced accuracy.

4. Out-of-fold prediction turns localisation into a fair downstream input

Five independently trained locators produce one joined mask set, but each patient is processed only by the locator that did not learn from that patient.

Cross-fitting: every patient receives a mask from a model that never trained on that patient

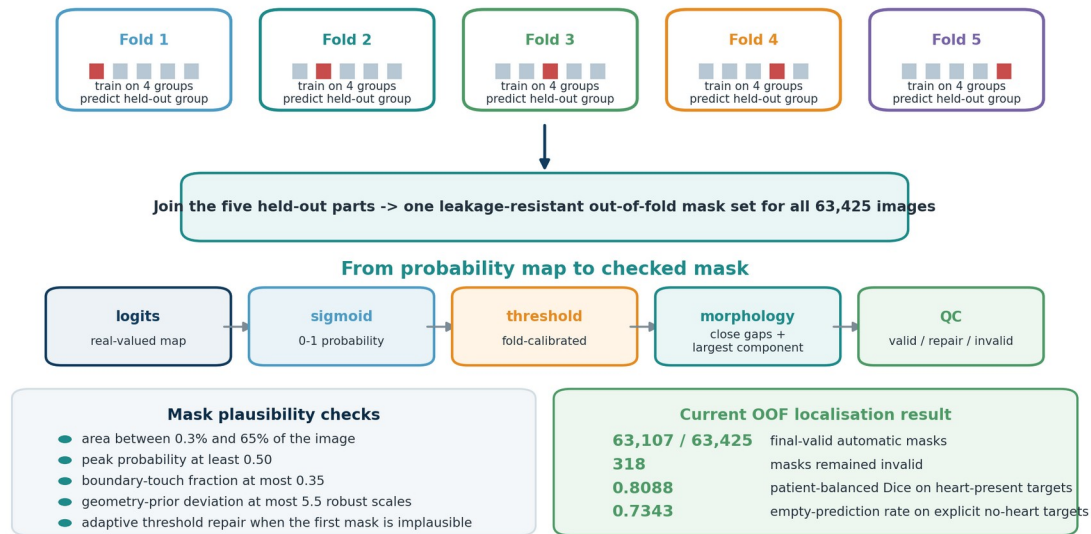


Figure 5. Patient-level cross-fitting, mask post-processing, plausibility checks, and current localisation metrics.

The raw network output is a probability map. A fold-specific threshold converts it to a binary mask; a 5 x 5 morphological closing fills small gaps; only the largest connected component is retained. Plausibility checks then consider mask area, peak probability, boundary contact, and deviation from a geometry prior learned from non-test masks. An implausible first mask can be repaired by trying nearby thresholds; unresolved cases remain invalid.

Output check	Current result / interpretation
Uncertainty signals	Pixel entropy, photometric test-time disagreement, heart-presence ambiguity, geometry deviation, and inconsistency with neighbouring frames.
Final-valid masks	63,107 of 63,425 released images; 318 remained invalid.
Positive-target quality	Patient-balanced Dice = 0.8088 on accepted heart-present targets.
No-heart failure mode	73.4% of explicit no-heart targets received the desired empty prediction; false-positive masks still remained on roughly one quarter.

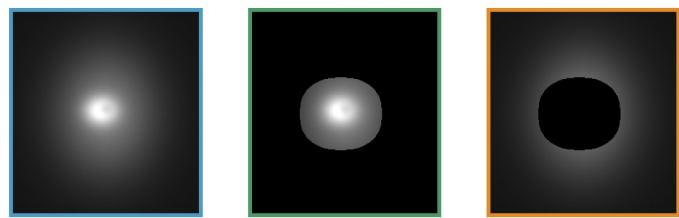
Why out-of-fold masks matter. If the heart locator had already trained on the test patient, its mask could encode patient-specific information. Cross-fitting makes the mask itself part of the held-out evaluation protocol.

5. The control experiment asks what the classifier can use

The strongest methodological step was not adding a larger classifier; it was constructing region controls from the same source slices.

The decisive experiment: three views of the same source slice

The 15 x 15 dilation retains a small context margin before ROI and complement are created.



B2 - full image AU1 - automatic ROI C1 - exact complement

can use heart + background forces attention toward heart tests residual shortcut signal

Cohorts and roles of the eleven representations

Cohort	Modes	Images	Scientific role
All released slices	B0	63,425	broad full-image reference
Primary same-slice	B2 / AU1 / C1	50,735	full vs ROI vs complement on identical slices
Secondary matched	B1 / AU2 / C2	17,750	8,875 Sick-Normal pairs; outcome-informed sensitivity analysis
Manual-positive	M1 / C3 / AU3 / C4	3,949	manual vs automatic ROI on the same annotated images

Why the complement matters: if pixels outside the heart still classify well, acquisition style or other dataset-specific clues remain available.

Figure 6. Full image, automatic cardiac ROI, and exact complement, followed by the four analysis cohorts and eleven fixed modes.

For the primary analysis, an image is eligible only when quality is valid, the final out-of-fold mask is valid, the heart-presence head is positive, and the mask covers at least 0.3% of the image. The same 50,735 slices and 2,541 series proxies are then used for B2, AU1, and C1. The automatic support is expanded with a 15 x 15 max-pooling operation so that the ROI includes a modest context margin; the complement is exactly the visible content outside that support.

Shortcut-learning logic. If AU1 is stronger than B2 and C1 on exactly the same slices, useful label-related information is more concentrated near the heart. If C1 remains above chance, however, information outside the intended region still correlates with the label.

Secondary controls

- Matched cohort: 8,875 Sick-Normal pairs (17,750 images) selected by appearance, geometry, sequence position, and quality; because labels are used to form pairs, this is a secondary sensitivity analysis.
- Manual-positive cohort: 3,949 identical annotated images compare manual ROI, automatic ROI, and their complements without changing the source images.

6. Frozen features and hierarchical pooling create one representation per patient

Every view is encoded by the same fixed image model, then averaged first within series and only then across series so that a long scan does not dominate.

Stage 3 - Final CPU: turn thousands of slices into one score per patient

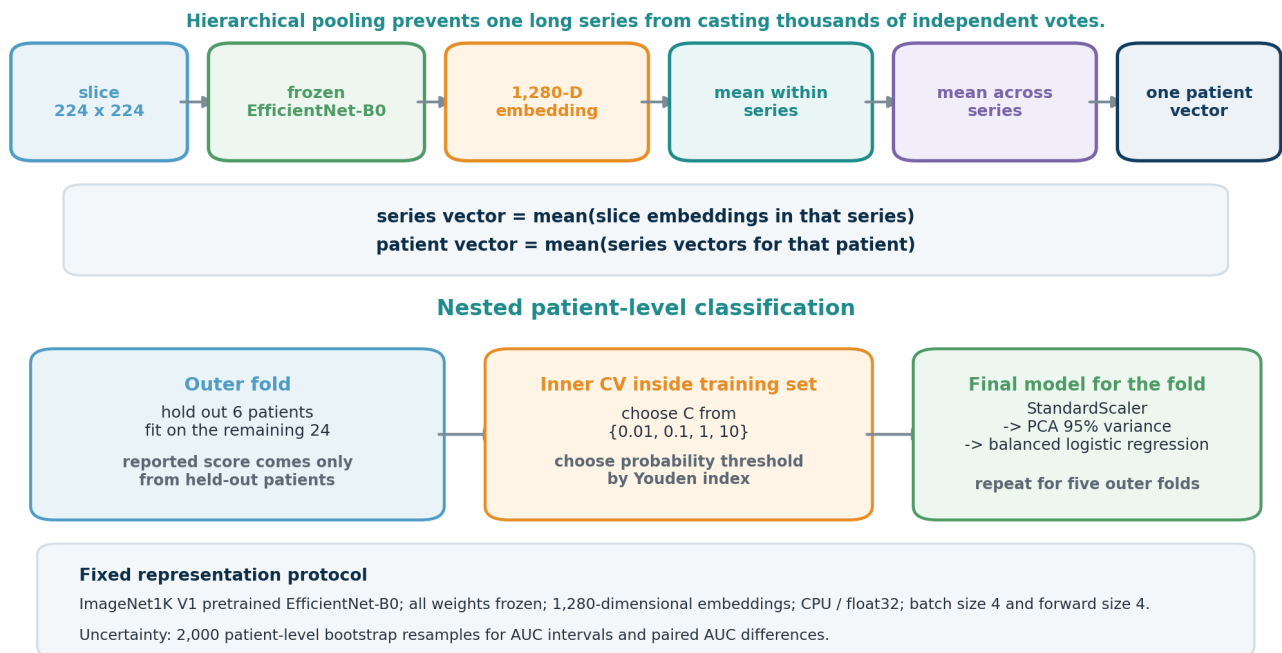


Figure 7. Frozen EfficientNet-B0 feature extraction, hierarchical pooling, and nested patient-level evaluation.

An ImageNet1K V1-pretrained EfficientNet-B0 converts each 224 x 224 view into a 1,280-dimensional embedding. All EfficientNet parameters are frozen: no parameter is updated with the CAD-related label. To preserve the defined numerical protocol, extraction runs on CPU in float32 with batch and forward size 4. Eleven representation banks are built from the cohorts defined on the previous page.

Evaluation layer	Implementation
Outer evaluation	Five stratified patient folds; each fold predicts six held-out patients.
Inner model selection	Within the outer training patients, choose logistic-regression C from {0.01, 0.1, 1, 10} and choose a decision threshold by the Youden index.
Final classifier	StandardScaler -> PCA retaining 95% variance -> class-balanced logistic regression.
Patient uncertainty	2,000 patient-level bootstrap resamples for AUC intervals and paired AUC differences.
Reported unit	One out-of-fold score per patient; no slice is treated as an independent vote.

Running example. The illustrative slice becomes one 1,280-number vector. It is averaged with other slices in its series; the series vector is averaged with the patient's other series; only the resulting patient vector enters the final classifier.

7. Results: the heart-focused view was strongest, but the sample remains small

All numbers below are patient-level out-of-fold results from the complete 5 October 2026 run with 5,097 accepted targets and complete recomputation of dependent stages.

Patient-level results from the complete 5 October 2026 run

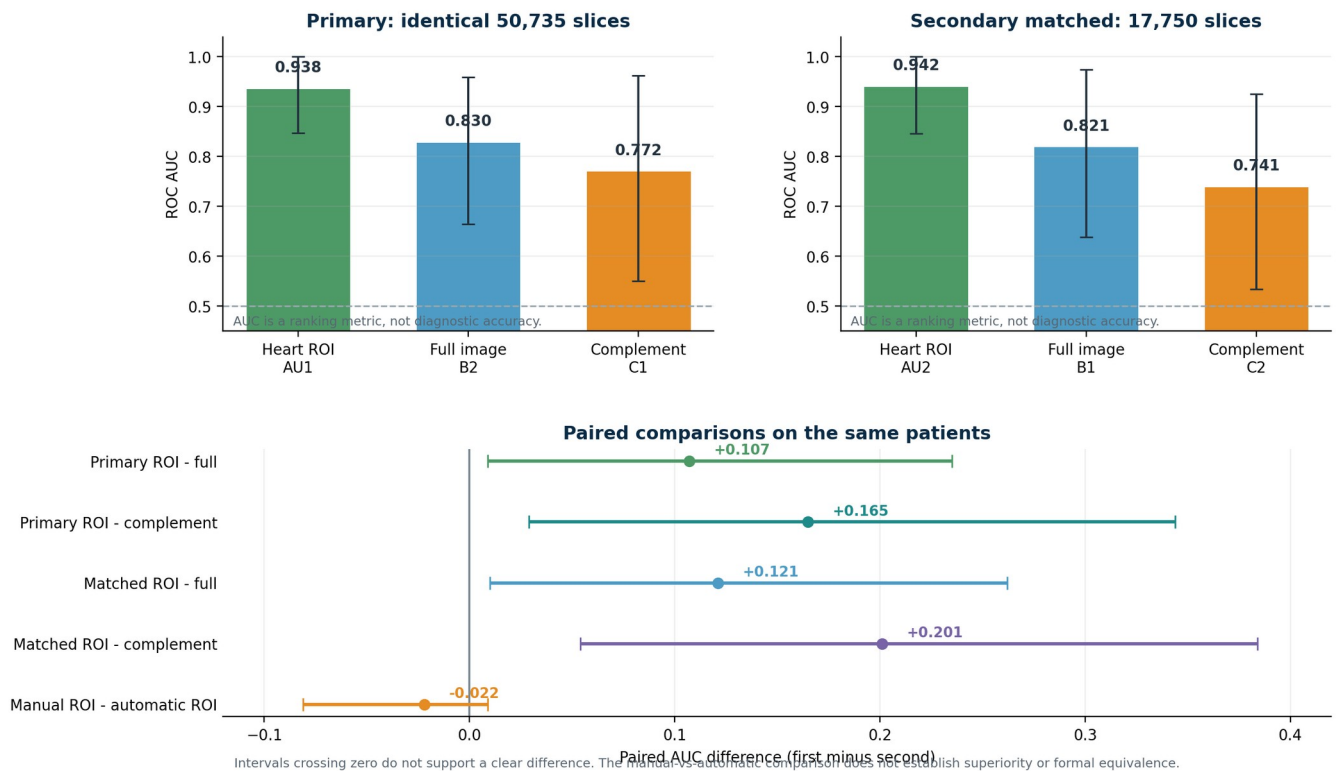


Figure 8. AUCs with 95% patient-bootstrap intervals and paired AUC differences. The primary comparison uses identical source slices.

Result	Interpretation
Primary automatic ROI (AU1)	AUC 0.938, 95% CI 0.847-1.000; AP 0.938; Brier 0.104.
Primary full image (B2)	AUC 0.830, 95% CI 0.665-0.959.
Primary complement (C1)	AUC 0.772, 95% CI 0.550-0.963.
Primary paired differences	AU1-B2 = +0.107 (0.009 to 0.235); AU1-C1 = +0.165 (0.029 to 0.344).
Manual-positive comparison	Automatic ROI AUC 0.973 vs manual ROI 0.951; M1-AU3 = -0.022 (-0.081 to 0.009), so no clear difference was established.

Best-supported interpretation. Within this release, label-related information is more concentrated in the out-of-fold cardiac region than in the full image or extra-cardiac complement. Complement AUCs of 0.741-0.835 still reveal substantial residual shortcut signal.

What is not supported. AUC 0.938 is not 93.8% diagnostic accuracy. The experiment does not prove that the model learned coronary disease, does not establish clinical usefulness, and does not show that automatic masks are anatomically superior to expert masks.

8. Reproducibility, contribution, limitations, and the next decisive test

The final story is not only that the model worked; it is that every claim is tied to a recoverable computation, an explicit control, and a stated boundary.

Reproducibility is part of the scientific argument

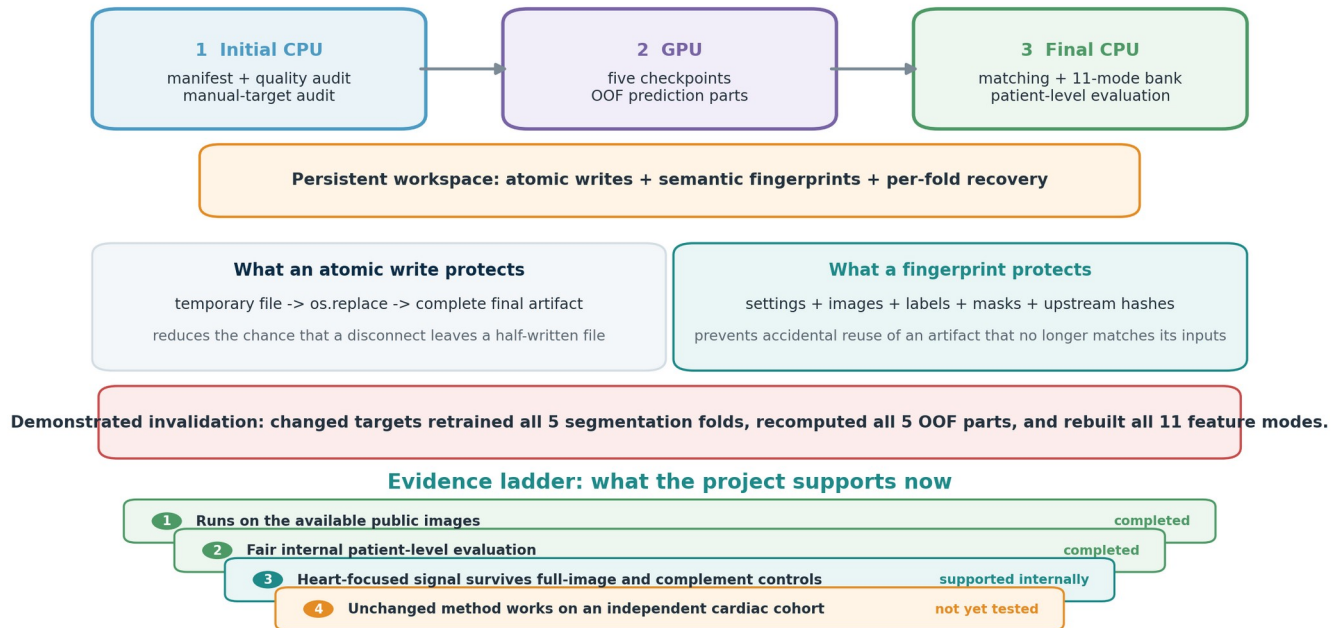


Figure 9. Three restart-safe Kaggle sessions, two distinct persistence safeguards, and the current evidence ladder.

The implementation separates the run into an initial CPU audit, a GPU segmentation stage, and a final CPU feature/evaluation stage. PNG, CSV, JSON, PyTorch, and NPZ artifacts are written atomically. Semantic fingerprints cover settings and upstream dependencies, so a file is reused only when its schema, expected content, and lineage remain compatible. In the reported run, changed annotations correctly forced every segmentation fold and prediction part to be recomputed, after which the changed feature fingerprint rebuilt all eleven representations.

Contribution layer	Documented role
David Vlad Marghidan	Dataset and folder audit; correspondence with the uploader; redesign around patient-level validation; pipeline implementation and refactoring; manual-mask review; full/ROI/complement controls; execution, analysis, figures, and writing.
Prof. Dr. Smaranda Belciug	Scientific supervision, methodological guidance, review of assumptions and conclusions, and research environment at INFusion AI Lab.
Generative-AI assistance	Used for code-review and refactoring discussions, debugging, documentation, language editing, consistency checking, and figure-layout preparation; it did not execute the Kaggle run or generate the reported metrics.

Principal limitations and next test

- Only 30 independent patients from one public source; confidence intervals are wide and may be unstable.
- JPEG exports rather than original DICOM; no verified sequence identifiers, scanner parameters, demographics, covariates, or independent re-adjudication of the supplied labels.
- Manual masks were created or reviewed by one non-radiologist researcher and were not independently adjudicated by a cardiac radiologist.
- The matched analysis is outcome-informed, and hyperparameters and quality gates were developed on the same release.
- The decisive next step is a frozen protocol tested on an independent cardiac DICOM cohort with radiologist-reviewed masks, full acquisition metadata, calibration analysis, and explicit shortcut controls.

Selected references

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Closing statement. The project moved from “Can a model obtain a high score?” to “What produced that score, which controls survive, and what evidence is still missing?” That methodological shift is the central research contribution.